

Veridian Corp — Internal IT Support Agent

System Architecture & Technical Design Specification | Assignment 2 Prototype

Target Organization: Veridian Corp	Coursework: B.Tech IT / Computer Science
Project Scope: Internal Service Agent (IT Triage)	Deployment: Local (Streamlit :8501) & Vercel (WASM)
Timeline: 21 Sep – 25 Sep 2026	Engine: Rule-Based, Policy-Grounded (Zero Hallucination)

1. Executive Summary & Core Objectives

The Veridian Corp Internal IT Support Agent is a deterministic, automated decision-support system engineered to triage and resolve incoming employee IT inquiries. Built strictly against the specifications of **Assignment 2: Internal Service Agent (IT Support)**, the agent provides immediate, policy-grounded resolutions while eliminating ticket misrouting and safeguarding sensitive corporate systems.

- **Strict Policy Grounding:** Exclusively applies KB-01 through KB-10 and the Asset Management Policy without external assumptions.
- **Zero Hallucination & Zero Cost:** Pure deterministic rule engine requiring no paid third-party API keys or external LLM tokens.
- **Precedent-Aware:** Directly links resolutions to historical ticketing records (TK-1042 to TK-1051) for organizational consistency.
- **Dual Runtime:** Operates natively via Python Streamlit locally, and via *stlite* WebAssembly on Vercel without servers.

2. High-Level System Architecture (Tiered Model)

Tier	Subsystem & Technologies	Core Function & Responsibilities
1. Presentation Tier	Streamlit Client / stlite WASM (3 Tabs: Agent, Requests, Tickets)	User interface for employee inquiry input, preset case selection, live decision badge rendering, and policy references.
2. Application & Triage Tier	app.py analyze_request() Engine	Performs tokenization, numerical threshold extraction (attempt counts, hardware age), vagueness filtering, and role routing.
3. Knowledge & Policy Tier	data.py POLICIES Dictionary	11 official corporate policies covering Password Resets, VPN, Hardware, Software, Spoolers, Mailboxes, Wi-Fi, Expenses, Security, and WFH.
4. Precedent & History Tier	data.py TICKET_QUEUE & Historical DB	10 ticket queue records distinguishing active operational routing from closed tickets that establish binding resolution precedents.

3. Repository & Module Structure

File	Type	Architectural Purpose
app.py	Python / Streamlit	Main application entrypoint, interactive triage interface, and rule classification engine.
data.py	Python Data Module	Static storage of 11 policies, 15 pre-computed employee requests, and 10 ticket records.
index.html	Static WebAssembly	Mounts stlite (Pyodide in WASM) to run app.py directly inside client browsers on Vercel.
vercel.json	Deployment Config	Configures clean URL rewrites and static asset routing for instantaneous Vercel Edge hosting.
requirements.txt	Dependencies	Declares streamlit as the single runtime requirement. Zero bloat.
README.md	Documentation	System manual, running instructions, limitations, and a structured viva examination defense guide.

4. Request Processing Pipeline & Evaluation Stages

- **Stage 1: Ingestion & Quality Gate:** Validates input length. Triggers 'Ask for More Information' on ambiguous queries (REQ-15).
- **Stage 2: Critical Security Triage (KB-09):** Scans for phishing/malware. Automatically flags quarantine order halting email forwarding (REQ-08).
- **Stage 3: Numerical Threshold Checks (KB-01):** Checks failed attempts. ≥ 5 attempts (REQ-03 with 6) routes to IT manual unlock; < 5 is self-service.
- **Stage 4: Role & Governance Boundary Verification:** Enforces contractor approval for VPN (KB-02, REQ-11) and directs expense tool accounts to Finance (KB-08, REQ-12).
- **Stage 5: Asset Refresh & Lifecycle Evaluation:** Cross-references KB-03 (3-yr eligibility) with 4-yr Asset Policy; routes 3.5-yr dead laptop (REQ-01) for Finance sign-off.
- **Stage 6: Precedent Association & Formulation:** Matches ticket queue history (TK-1042..1051) and renders Category, Policy, Decision, Action, and Reason.

5. Standardized Decision Taxonomy

Decision Type	Operational Meaning	Assignment Example
Resolve / Self-Service	Handled by employee via portal/kiosk without ticket.	REQ-02 (Guest Wi-Fi), REQ-05 (VPN renew)
Route to IT	Direct manual action by IT technician required.	REQ-01 (3.5 yr dead laptop), REQ-03 (6 password attempts)
Route to Security	Immediate incident reporting and threat quarantine.	REQ-08 (Phishing email being forwarded)
Route to Finance	Budget approval, allowance processing, or tool creation.	REQ-07 (WFH monitor allowance), REQ-12 (Expense tool)
Manager Approval Required	Requires formal business justification and manager sign-off.	REQ-10 (Admin server access), REQ-11 (Contractor VPN)
Security Review Required	IT Security software vetting (takes 3–5 business days).	REQ-04 (Non-catalog tool), REQ-14 (Browser extension)
Troubleshooting Required	Hardware diagnostics or spooler service restart.	REQ-06 (Printer jam), REQ-13 (2-yr flickering screen)
Ask for More Information	Clarification requested; input lacks actionable specifics.	REQ-15 ('hey can you help, its not working')

6. Ticket Queue Architecture & Precedent Linkage

Section 3 of the assignment contains 10 ticket records. The system utilizes them systematically:

- **Active Work Items (4 tickets):** TK-1043 (Laptop 3.2 yrs), TK-1044 (Non-catalog software), TK-1047 (Home office equipment), and TK-1048 (Phishing) represent active cases pending fulfillment or external review.
- **Historical Precedents (6 tickets):** Closed tickets govern consistency. Specifically, **TK-1050** sets the organizational precedent that administrative server access without business justification must be rejected.

7. Dual-Mode Deployment Topology

Dimension	Local Execution (Streamlit)	Cloud Serverless Execution (Vercel)
Target Runtime	Python 3.11 Local Interpreter	Pyodide / WebAssembly in Browser (stlite)
Serving Model	Uvicorn / WebSocket server on localhost:8501	Edge Static CDN Delivery (Zero Server Overhead)
Launch Trigger	python -m streamlit run app.py	Git commit to GitHub repository
Operational Role	Live examiner evaluation & viva defense	Public shareable link for AIONOS reviewers

8. Corporate Governance & Security Constraints

- **Auditable Decision Traceability:** Each resolution provides an unambiguous policy citation (e.g., KB-01), creating an immutable, reviewable audit trail for corporate IT compliance.
- **Strict Segregation of Duties (SoD):** Adheres to corporate authorization boundaries. IT technicians cannot grant expense software access (Finance jurisdiction under KB-08) or bypass security reviews (KB-04).
- **Active Incident Containment:** Security threats flagged under KB-09 trigger explicit warnings preventing employees from broadcasting phishing lures internally.

Project: Veridian Corp IT Support Agent	Author: B.Tech Computer Science / IT Student
Repository: github.com/kisna69/internal-service-agent	Status: Fully Tested & Verified (All 15 Requests Passed)